

FEDSHIELD-ID

Privacy-First Identity Trust Platform for Banking Networks

PROJECT OVERVIEW

A comprehensive overview of FedShield-ID, detailing the system goals, capabilities, security architecture, and operational frameworks for privacy-preserving user and threat analytics in modern retail and savings banking networks.

Prepared For: Security & Compliance Audit Team

Platform Version: v1.0.0 (Production Release)

Classification: Banking Network Confidential

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1. Executive Summary

FedShield-ID is an enterprise-grade, privacy-first identity trust and verification platform designed specifically for modern banking networks. In an era where digital financial transaction volumes are surging, banking networks face sophisticated, coordinated fraud vectors including synthetic identities, device reputation spoofing, credential stuffing, SIM swap takeovers, and insider collusion.

To mitigate these vulnerabilities without imposing friction on trusted customers, FedShield-ID continuously evaluates customer and employee identity trustworthiness. Rather than scoring transactions in isolation, the platform dynamically computes a multi-dimensional Identity Trust Score using behavioral biometrics, device telemetry, geo-distance drift, and historic recovery logs. When elevated risk is detected, the system triggers granular Adaptive Authentication challenges (ranging from SMS OTP to live Face Verification), blocking malicious activities in real-time.

Critically, FedShield-ID utilizes Privacy-Preserving Federated Learning and Laplacian Differential Privacy to share threat intelligence across distinct banking entities without pooling sensitive customer database entries. Additionally, it is fortified with Crystals-Kyber-768 Post-Quantum cryptography to secure data streams against future quantum computing threats, and utilizes SHAP-based Explainable AI to render compliance-ready audit reports.

2. Core Product Modules

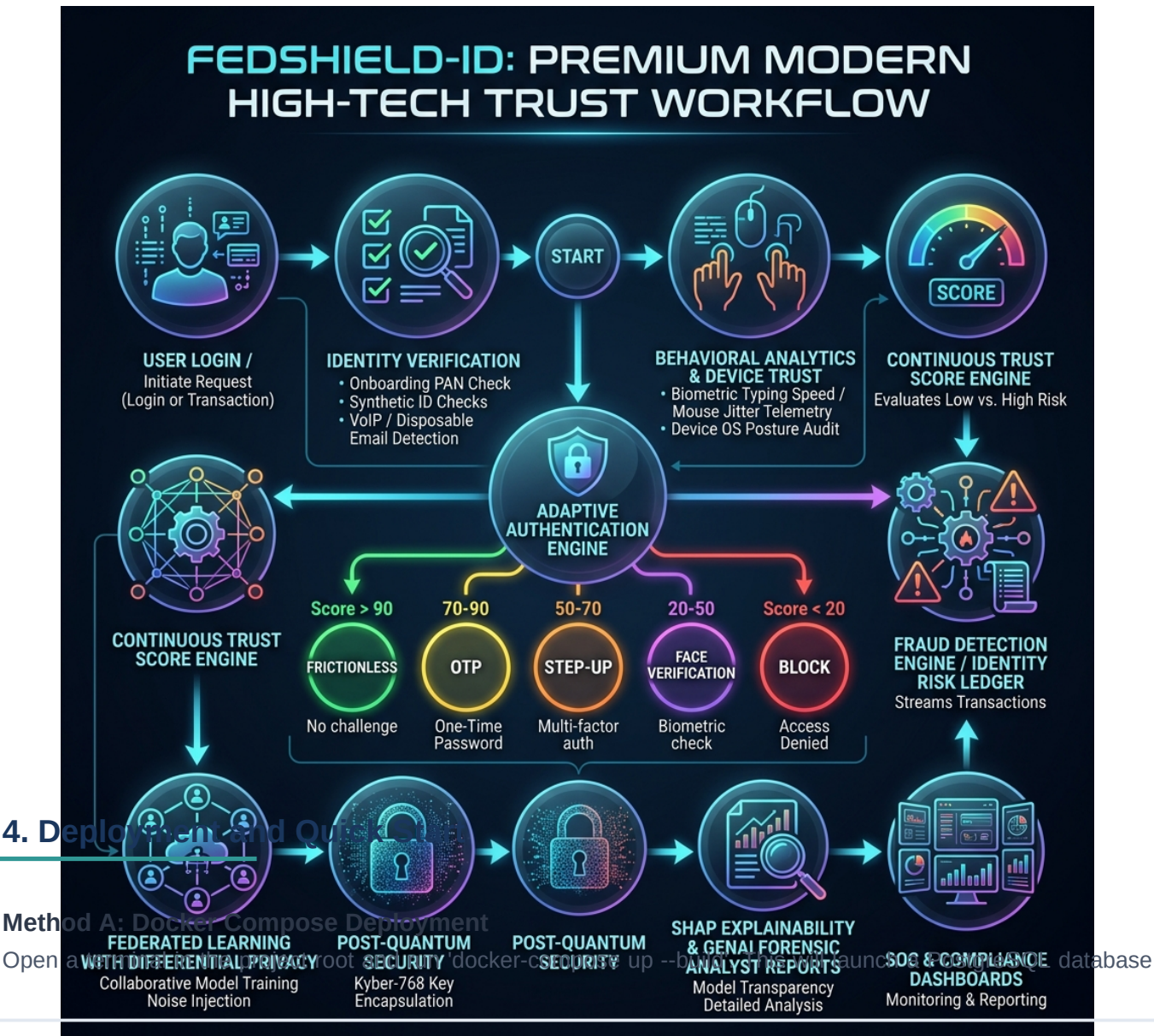
The platform organizes its security defenses around twelve core product modules designed to address the entire identity lifecycle from onboarding to recovery:

- * **Onboarding Verification and Synthetic ID Prevention:** Validates PAN formats, flags tempmail domains, and identifies VoIP carriers to block synthetic accounts at inception.
- * **Continuous Identity Trust Scoring:** Dynamically updates user trust categories based on behavioral telemetry, volume spikes, and geo-travel anomalies.
- * **Behavioral Biometrics Telemetry:** Captures millisecond-level typing speed, mouse movements, and click latency to establish unique human signatures.
- * **Device Trust Intelligence:** Audits operating system parameters (rooted, jailbroken, or emulated status) and device network reputations.
- * **Adaptive Authentication Engine:** Applies dynamic, score-based verification paths, enforcing Step-Up auth or hard blocks depending on real-time threat levels.
- * **SIM Swap and Recovery Telemetry:** Audits account recoveries requested within 72 hours of a mobile carrier SIM swap, preventing Account Takeover (ATO).
- * **Insider Threat Monitoring:** Audits privileged administrator actions, flagging database queries performed out-of-hours or large customer data exports.

- * **Privacy-Preserving Federated Learning:** Collaboratively trains machine learning classifiers across three nodes (Bank A: Retail, Bank B: Cards, Bank C: Savings) without centralizing sensitive customer records.
- * **Differential Privacy (DP):** Injects Laplacian noise into local ML weights before federation, mathematically ensuring that individual records cannot be reconstructed.
- * **Post-Quantum Cryptography (PQC):** Simulates CRYSTALS-Kyber-768 Key Encapsulation Mechanism (KEM) to establish quantum-safe encryption tunnels for model weights.
- * **Explainable AI (SHAP):** Exposes model decision attributes using Shapley Values, rendering clear visual explanations of risk contributors.
- * **Identity Risk Ledger and Graph:** Streams transactions in real-time to a local ledger and builds a collision graph to expose money laundering rings sharing hardware IDs.

3. System Workflow and Process Flow

Below is the continuous trust verification pipeline mapping customer interactions from initial login through the real-time scoring backend, post-quantum tunnels, federated modeling, and SOC dashboard alerts:



along with the FastAPI backend and Vite/React frontend dashboard. The React client is accessible at <http://localhost:3000>.

Method B: Manual Local Setup (SQLite fallback)

1. Backend: Navigate to 'backend/', initialize virtual environment, run 'pip install -r requirements.txt', and launch server using 'python -m uvicorn app.main:app --reload'.
2. Frontend: Navigate to 'frontend/', run 'npm install' and 'npm run dev' to boot the Vite development server.